
EE/CprE/SE 492 WEEKLY REPORT 3

September 18, 2025 - October 2, 2025

Group number: 16

Project title: Outdoor Roomba

Client &/Advisor: Prof. Diane Rover

Team Members/Role:

Brodie M Bates - Embedded system

Daniel Vergara Pinilla - Machine Learning

Mitchell D Owen - Project Tester

Om Shukla - Frontend Developer

Sudipta Halder - Simulation developer

○ Weekly Summary

This week on the hardware side we printed motor mounts and put them onto the base of the robot. Then, we tested this by running just one and then connected both and made the robot move. Also, we set up a microcontroller to be able to drive the motors with a given serial connection.

○ **Past weeks accomplishments**

- Sudipta Halder: Got the robot to move and printed more parts.
- Brodie Bates: Got the ESP32 to be able to control the motors based on a signal from a serial connection. Worked on the LiDAR code more, set up the battery board, tested motors while mounted to the chassis.
- Daniel Pinilla: Improved accuracy of the model as well as working on the map visualization for roomba
- Om Shukla: Working on sending requests to the backend for user authentication. Currently running into an issue with CORS (Cross-Origin Resource Sharing) that is not allowing requests to the backend. I'm Implemented a temporary workaround using my local server as a proxy request server to handle backend requests, and I am working on implementing a solution that solves the CORS issue.
- Mitchell Owen: putting the motor brackets on the robot and finally getting the robot to move.

- **Pending issues**

- Om Shukla: Fix the CORS issue with the backend not allowing requests from the frontend.

○ **Individual contributions**

<u>NAME</u>	<u>Individual Contributions</u>	<u>Hours this week</u>	<u>HOURS cumulative</u>
Brodie Bates	Motor controller from serial connection code, Battery controller soldering, assisting in running motors and testing designs.	14	23
Om Shukla	Working on requests to the backend for user authentication.	5	17
Sudipta Halder	Got the robot to move and printed more parts.	6	31
Daniel Mauricio Vergara	Manage the websocket communication with the new map on the app	4	16
Mitchell Owen	Assembly of the chassis of the robot and testing the motors.	4	13

○ **Plans for the upcoming week**

- Sudipta Halder: Continue printing parts.
- Brodie Bates: Finish LiDAR code, make a mount for the front set of wheels, get robot moving.
- Om Shukla: Finish, if not then continue working on a implementation for the backend to resolve the CORS issue with the request handling.
- Mitchell Owen: Catching up on what I missed since my surgery. Reevaluating what I can do for the time while I recover.
- Daniel Pinilla: Manage accuracy on the map imagining on the front end

- **Summary of weekly advisor meeting**

The first week of this report we had an advisor meeting where we showed the motor controller and input. Our advisor asked us about the wooden base of the robot and mentioned how it should be changed in the future since wood is a heavier material. Also, our advisor asked if we were updating our CAD model. We also talked with our advisor about our pickup mechanism and how we should simulate how much we can pick up weight wise for a performance requirement. Another thing we mentioned in the meeting is our plan for our sensors and that we should look at similar robots and base our materials off of what they use.